

Comprehensive PYQ Analysis & Exam Blueprint Report

Basic Computer Engineering (BT-205 GS)

University: Rajiv Gandhi Proudyogiki Vishwavidyalaya (RGPV), MP

Course Code: BT-205 (GS)

Analysis Horizon: 2022 – 2025 (8 Exam Cycles)

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Section 1: Unit-Wise Previous Year Questions (PYQs)

This section compiles every question extracted from the 8 consecutive semesters (2022–2025) without omission, sorted logically into standard RGPV syllabus units.

Unit I: Computer Hardware & Software Fundamentals

Define computer and explain its classification with examples. [Dec 2025](#) [Dec 2024](#) [June 2023](#) [Nov 2022](#)

Describe CPU organization including registers, bus architecture and instruction set. [Dec 2025](#)

Explain memory and storage systems along with I/O devices. [Dec 2025](#) [June 2023](#)

Discuss system software and application software with examples and explain what distinguishes them from one another. [Dec 2025](#) [Dec 2024](#) [Dec 2023](#)

Classify computers based on size and functionality. Compare supercomputers, mainframes and embedded systems. [June 2025](#) [June 2023](#)

Explain the architecture and working of a CPU with a labeled diagram. [June 2025](#)

Define computer memory and differentiate between primary and secondary memory. [Dec 2024](#)

What are the advantages of a computer? Discuss with suitable examples. [Dec 2024](#)

Explain the organization of computer with help of block diagram. [June 2024](#)

Give examples of basic mathematical functions in Excel such as SUM, AVERAGE and MAX. [June 2024](#)

How do you insert headers and footers into a Word document? What content can be included in headers and footers? Discuss. [June 2024](#)

What is Microsoft Office and what is its primary purpose? [June 2024](#)

Discuss the generation of computers. [June 2024](#)

Could you explain the various memory and storage techniques typically utilized by computers. **Dec 2023**

Explain I/O devices in detail with suitable examples. **June 2023** **June 2024**

Explain different computer memories in RAM, ROM, CACHE, REGISTER and HARD DISK in detail. **June 2022**

Write short notes on: BUS Architecture. **Nov 2022**

Write short notes on: MS Word Vs MS Excel. **Nov 2022**

Unit II: Operating System Concepts & Program Design

Explain operating systems and its main functions / key responsibilities. **Dec 2025** **June 2023** **June 2022**
Nov 2022

Discuss file, process and memory management in operating systems. **Dec 2025**

Explain algorithms, flowcharts and program design. **Dec 2025**

Define an Operating System. Discuss its structure and explain the concept of a kernel. **June 2025**

What is an algorithm? Write an algorithm to find largest value from the given three numbers. **June 2025**

What are limitations do flowcharts have in representing complex algorithms or processes? **Dec 2024**

Explain how to create a flowchart for a simple algorithm, such as finding the maximum of two numbers.
Dec 2024

What is an operating system, and what are its key responsibilities in a computer system? Also, discuss various types of operating systems now in use. **Dec 2023**

Discuss the various sorts of algorithmic complexities, such as time and space complexity, as well as how they are quantified. **Dec 2023**

Define algorithms. What is the need of algorithms? Describe three benefits of algorithms. **June 2023**

Why Algorithm writing and drawing flow chart is necessary before writing a computer program? **Nov 2022**

Unit III: Object-Oriented Programming & C++ Basics

Discuss programming paradigms and characteristics / features of OOP.

Dec 2025

June 2025

June 2023

June 2022

Compare procedure oriented programming with object oriented programming / functional programming.

Dec 2025

Dec 2023

June 2023

Explain basic structure of C++ program and its tokens.

Dec 2025

Explain arrays and functions in C++.

Dec 2025

Discuss constructors, destructors and scope resolution operator.

Dec 2025

List and explain various operators in object oriented program.

June 2025

Write a C++ program to swap two numbers.

June 2025

How many types of constructors are present in C++? Explain in detail.

June 2025

Demonstrate the working of a friend function in C++ with an example / program.

June 2025

Dec 2023

June 2022

Write a program to read and print employee information using multiple inheritance.

June 2025

What is a class in C++ and how does it differ from an object? Provide an example.

Dec 2024

Dec 2023

June 2023

Explain the concept of constructors and destructors in C++.

Dec 2024

June 2022

What is encapsulation and why is it important in OOP?

Dec 2024

Write short note: Logical operators in C++.

Dec 2024

What are the principles of Object-Oriented Programming (OOP)?

June 2024

What is the difference between single inheritance and multiple inheritance?

June 2024

Explain the concept of operator overloading with an example.

June 2024

What is operator overloading and why is it useful in C++? **June 2024**

Explain the difference between one-dimensional and multi-dimensional arrays. **June 2024**

Compare and contrast low-level programming languages (e.g., assembly language) and high-level programming languages (e.g., Python) in terms of their characteristics and use cases. **Dec 2023** **Nov 2022**

Explain what "data types" are used in C++ and what the difference is between "primitive" and "derived" data types? Give an example of each one. **Dec 2023**

Explain the inheritance in object-oriented programming and its implications in terms of code reuse and extensibility. Discuss the various forms of inheritance, such as single inheritance, multiple inheritance and multilevel inheritance. **Dec 2023** **June 2022**

Explain the following: data type, tokens, variables, operator. **June 2023**

Define objects and classes. Can a class in C++ have more than one constructor with the same name? Justify your answer with suitable example. **June 2023**

Explain Function overloading and Operator overloading with an example. **June 2023** **Nov 2022**

Explain the various kinds of looping statements in C++? With examples. **June 2022**

Write a program in C++ to convert the given binary number into the decimal number. **June 2022**

How Function overloading and operator overloading is done? Explain them with a simple program. **Nov 2022**

Write a C++ program to insert and delete an element from stack. **Nov 2022**

What is an Array? Explain different types of Arrays with syntax and suitable example programs. **Nov 2022**

Write short notes on: Polymorphism. **Nov 2022**

Unit IV: Computer Networks & Cyber Security

Explain ISO-OSI model and functions of its layers. **Dec 2025** **Dec 2024** **June 2022**

Discuss TCP/IP model, internetworking devices and Internet applications. **Dec 2025**

Differentiate between TCP/IP and ISO-OSI reference model. [June 2025](#) [Dec 2023](#) [Nov 2022](#)

What is a Firewall? Illustrate the working principle of a firewall. [June 2025](#) [Nov 2022](#)

Explain the procedure involved in protecting devices from malware, virus and worms. [June 2025](#)

What is the TCP/IP model and why is it significant in computer networking? [Dec 2024](#)

Explain common types of cyber threats, such as malware, phishing and Denial-of-Service (DoS) attacks. [June 2024](#)

Examine several types of malware such as viruses, worms, trojans and spyware. Examine their properties, mechanisms of propagation and potential effects on computer systems. [Dec 2023](#)

Determine internetworking and the difficulties associated with joining numerous networks. Discuss the importance of technologies such as routers and switches. [Dec 2023](#)

Explain TCP/IP model in detail with a diagram. [June 2023](#) [June 2022](#)

Describe Denial of Service (DoS) and logic bombs in detail. [June 2023](#) [June 2022](#) [Nov 2022](#)

Write a short note on good computer security habits. [June 2023](#) [June 2022](#)

Write short notes on: E-mail Spoofing. [June 2022](#)

What is E-Commerce? Write the advantages and disadvantages. [Nov 2022](#)

Unit V: Database Management Systems & Advanced Computing

Explain DBMS and its advantages over file system. [Dec 2025](#)

Discuss cloud computing models, types and advantages. [Dec 2025](#)

Describe the functions of a DBA in maintaining the performance, integrity and security of a database. [June 2025](#) [June 2023](#) [Nov 2022](#)

Discuss about the different cloud service delivery models. [June 2025](#) [Dec 2024](#)

Describe the key components of a typical DBMS architecture. [Dec 2024](#)

What is cloud computing and how does it differ from traditional computing models? **Dec 2024**

Write short note: Data Independence. **Dec 2024** **June 2023**

What is a database transaction and why is it important in maintaining data integrity? **June 2024**

Write short note: DDL and DML Commands. **June 2024** **Nov 2022**

Explain the following with a suitable example: Data dictionary. **Dec 2023**

Describe data independence and DBA in detail. **June 2023**

What do you understand by cloud computing? Describe cloud infrastructure in detail? **June 2023** **June 2025**
Dec 2023 **Nov 2022**

Write short note: Pros and cons of cloud computing. **June 2023** **June 2022**

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Draw and Explain the architecture database systems. **June 2022**

Write short note: Data Manipulation Language. **June 2022**

Unclassified Cross-Disciplinary Questions

Note: These questions utilize computing concepts but belong primarily to domain applications (e.g., E-Business, GIS, Law) or are mixed-domain items found in standard papers.

Explain the applications of computer in the field of Meteorology and Climatology, Multimedia and Animation. **June 2022** **June 2025**

Explain the elements required to prove internet fraud under the law. **Dec 2024**

Write short notes on: Cyberstalking. **June 2025**

Describe the following: Remote sensing and GIS. **June 2023** **June 2024**

Describe the following: Computer applications in E-business / role of social media in e-commerce. **June 2023**
June 2024

Section 2: Important Topics & Priority Matrix

The following table tracks high-priority parameters, structural definitions, and concept paradigms across all core units.

Unit	Topic / Concept Breakdown	Priority	Exam Relevance / Reasoning
Unit I	Computer Classification (Size/Functionality); System vs Application Software differences; CPU Block Diagrams.	★★★★ High	Forms the baseline architectural theory requested across major descriptive cycles.
Unit II	Operating System Functions (Process, File, Memory); Algorithm design & Flowchart structures.	★★★★ High	Consistently tested via programmatic design questions or numerical algorithms.
Unit III	OOP Characteristics (Encapsulation, Polymorphism, Inheritance); C++ Constructors/Destructors, Friend Functions.	★★★★ High	Heaviest unit in terms of point weight, specific programs, and structural paradigms.
Unit IV	ISO-OSI 7-Layer model details; TCP/IP Model mapping & comparison; Firewalls and Malware types.	★★★★ High	Core data communication question present in every exam cycle.
Unit V	Cloud Service Models (IaaS, PaaS, SaaS); DBMS vs File System advantages; DBA Functions & Commands.	★★★★ High	Consistently maps out the architectural modern-computing section of the paper.
Unit III	Operator Overloading vs Function Overloading; Array structures (1D vs Multi-D).	★★★ Medium	Frequent secondary descriptive or programmatic choice.
Unit V	Data Independence; DBMS Architecture; DDL & DML operations.	★★★ Medium	Standard architectural component requirements.
Unit I	Excel Mathematical functions; MS Word formatting utilities.	★ Low	Found occasionally inside practical-application subparts.

Section 3: Most Repeated PYQs & Trend Analysis

Identical and closely mapped core question trends matching key parameters across the 8 exam cycles.

1. Characteristics, Principles, and Features of Object-Oriented Programming (OOP)

- **Asked in:** Dec 2025, June 2025, June 2024, June 2023, June 2022
- **Frequency:** 5 times

2. Cloud Computing Infrastructure, Service Models (IaaS, PaaS, SaaS), and Delivery Models

- **Asked in:** Dec 2025, June 2025, Dec 2024, Dec 2023, June 2023, June 2022, Nov 2022
- **Frequency:** 7 times (Highest recurring infrastructure topic)

3. Functions and Core Responsibilities of an Operating System (OS)

- **Asked in:** Dec 2025, Dec 2023, June 2023, June 2022, Nov 2022
- **Frequency:** 5 times

4. Comparison & Architectural Layer Mapping: ISO-OSI Model vs TCP/IP Model

- **Asked in:** Dec 2025, June 2025, Dec 2024, Dec 2023, June 2023, June 2022, Nov 2022
- **Frequency:** 7 times

5. Working Principles, Features, and Implementations of Constructors and Destructors in C++

- **Asked in:** Dec 2025, June 2025, Dec 2024, June 2022
- **Frequency:** 4 times

6. System Software vs Application Software Categorization

- **Asked in:** Dec 2025, Dec 2024, Dec 2023
- **Frequency:** 3 times

7. Role and Responsibilities of a Database Administrator (DBA)

- **Asked in:** June 2025, June 2023, Nov 2022
- **Frequency:** 3 times